

Evaluating Regression Techniques to Predict Housing Prices in Ames, Iowa

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MSDS-422 Summer 2025

Northwestern University, Practical Machine Learning

July 13, 2025

1. Introduction

What impacts house prices in Ames, Iowa, the most, and which model will allow us to predict these prices most accurately? Through this analysis, we seek to provide insights into the complex relationships that drive residential property values and support informed decision-making for multiple stakeholders.

2. Methodology

We analyzed 1,460 house sales from the “House Prices: Advanced Regression Techniques” dataset (Montoya & DataCanary, 2016). We discovered that lower-priced houses dominated the distribution and elected to reduce its skew through a log transformation normalization. Further, we filled in missing data points through median imputation and created new features such as `TotalSF` (sum of basement and ground living areas), `TotalBathrooms`, `HasFireplace` (a ‘0 and 1’ binary indicator), `IsModernHome` (built after 1980), and `TotalSF\_over\_2000` (identifies large homes) to capture perceived value.

To prepare our data for regression modeling, we selected the top 20 features with the strongest relationships to sale prices and performed a polynomial transformation to capture interactions between these features (e.g., bathrooms multiplied by fireplaces). Further, we standardized these features to ensure even scaling.

We selected four regression models for our analysis: (1) Linear Regression to act as an interpretable baseline, (2) Ridge Regression ($\alpha=100$) to reduce multicollinearity and prevent overfitting, (3) Lasso Regression ($\alpha=0.001$) to shrink large coefficients and auto-select features, and (4) ElasticNet Regression ($\alpha=0.01$, $l1_ratio=0.1$) to derive the benefits of both Ridge and Lasso penalties. We used bootstrap resampling, residual analysis, cross-validation, and other statistical tests to assess these models.

3. Analysis & Insights

Our analysis demonstrated the value that regularization brings to modeling complex scenarios as the regularized models beat linear regression (LR) by explaining 87% of the price variation. Further, our Ridge model consistently explained 86% of the price variation over a thousand resamples. However, we noted that our models priced low-to-mid tier houses well but underpriced higher-tier houses, which was confirmed by the left skew in their residuals.

Further, our Ridge and ElasticNet models performed better (through effective coefficient shrinkage) than LR (overfit to extreme coefficients) and Lasso (eliminated too many variables). Since Ridge is more balanced in its regularization and retains all features, it is more interpretable and relevant to stakeholders, making it our preferred model for this use case.

In terms of feature importance, we discovered that house prices are impacted by combinations of features, such as size and quality interactions, demonstrating the value of livability and reliability. We also noted that older houses seem to overcome their age disadvantages through strategic renovations, though newer houses command higher prices. Interestingly, homes beyond 2,000 square feet of living space did not have in-line price increases, especially when they lacked more modern construction or higher quality ratings.

4. Conclusion

Our findings demonstrate that house prices in Ames, Iowa, are driven by features by interacting with each other than individual characteristics. We also provided buyers, sellers, and policymakers with a predictive model to make more informed assessments, despite our data covering a narrow timeframe and lacking location-specific detail. Our model offers a solid foundation for extensions to broader markets and improved, data-driven pricing in real estate.

References

Montoya, Anna and DataCanary. 2016. “House Prices - Advanced Regression Techniques.”
Kaggle. <https://kaggle.com/competitions/house-prices-advanced-regression-techniques>